

Module:	Data Analytics
Assignment No:	2: Data Analytics Using KDD Process
Hand-Out Date:	Monday 7 th October 2024
Hand-In Date:	Friday 6 th December 2024
Lecturer:	Andrew Shields (andrew.shields@staff.ittralee.ie)

Requirements

Working individually, you are required to analyse a dataset employing the KDD process. Utilise Python, Jupyter Notebooks, Scikit-Learn, Pandas, or other data analysis tools to demonstrate your capabilities in data analytics.

Quality Metrics

The primary focus is on the quality of your work, which will be assessed based on the following criteria:

- Depth of analysis
- Rigour in following the KDD process
- Proper application of theories and methodologies
- Degree of originality

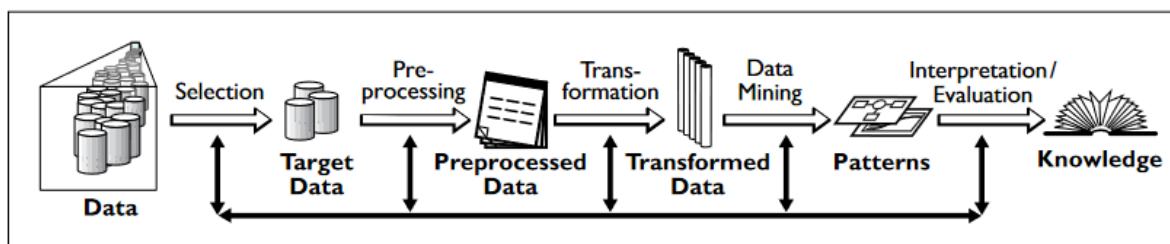
Research Element

A minimum of four academic papers must be reviewed to ensure that your analysis is rooted in existing research. Agree on the research topic and the 'theme' (title, thesis, hypothesis) in advance with your lecturer. The project should be both applied ("action-oriented") and theoretical.

KDD Process Example

For clarity, here is a breakdown of the KDD process stages you are expected to follow:

Figure 1. Overview of the steps constituting the KDD process



1. **Data Cleaning:** Removing noise and irrelevant data.
2. **Data Integration:** Combining data from multiple sources.
3. **Data Selection:** Selecting the data relevant for analysis.
4. **Data Transformation:** Transforming the data into a suitable format or structure.
5. **Data Mining:** Employing techniques to extract patterns.
6. **Pattern Evaluation:** Identifying truly interesting patterns representing knowledge.
7. **Knowledge Presentation:** Visualization, transformation, and the removal of redundant patterns.

Deliverables:

Notebook and Files

Submit all practical files, including a Jupyter Notebook, on Google CoLab or GitHub. Ensure that your notebook includes:

- A section for each stage of the KDD process
- Markdown content detailing your analysis results and experiments

Referencing

All references should follow the prescribed MTU Harvard style.

Grading Rubric

Total Marks: 100

- **Active Engagement:** 10 points
 - Criteria: Level of participation in the development of the assignment.
- **Technical Setup:** 10 points
 - Criteria: Efficiency in setting up the data analytics environment.
- **KDD Process:** 40 points
 - Criteria: Adherence to the Knowledge Discovery in Databases (KDD) process.
- **Broad Context:** 10 points
 - Criteria: Understanding of broader applications of used technologies.
- **Analytical Rigour:** 10 points
 - Criteria: Depth of analysis and validity of conclusions.
- **Reflection:** 10 points
 - Criteria: Depth of reflection, supported by academic references.
- **Scientific Method:** 5 points
 - Criteria: Application of the scientific method to data analysis.
- **Working Files:** 5 points
 - Criteria: Submission of all necessary working files.

Plagiarism

Plagiarism will be detected using the institute's plagiarism detection software and will be treated in accordance with institute policy.

Data Analytics Assignment Grading Rubric

Total Marks: 100

Active Engagement (10 points)

Criteria	Excellent (9-10 points)	Good (7-8 points)	Satisfactory (5-6 points)	Unsatisfactory (0-4 points)
Participation	Consistent and meaningful participation in the development of the assignment.	Regular participation with some notable contributions.	Occasional participation, but generally passive.	Little to no participation in the development of the assignment.

Technical Setup (10 points)

Criteria	Excellent (9-10 points)	Good (7-8 points)	Satisfactory (5-6 points)	Unsatisfactory (0-4 points)
Environment Configuration	Environment is set up flawlessly with all required packages and dependencies.	Environment is mostly set up correctly; minor issues that don't impact analysis.	Some issues in setup that require intervention but are eventually resolved.	Incorrect or incomplete setup affecting project outcome.

KDD Process (40 points)

Criteria	Excellent (36-40 points)	Good (28-35 points)	Satisfactory (20-27 points)	Unsatisfactory (0-19 points)
Adherence to KDD Steps	Excellent understanding and application of all stages of the KDD process.	Good application but misses one or two minor steps.	Moderate application but misses several steps or lacks depth.	Poor application; significant gaps in understanding or execution.

Broad Context (10 points)

Criteria	Excellent (9-10 points)	Good (7-8 points)	Satisfactory (5-6 points)	Unsatisfactory (0-4 points)
Application Insights	Demonstrates a comprehensive understanding of the technologies' broader applications.	Shows a good understanding with minor lapses.	General understanding with significant gaps.	Little to no understanding of broader application.

Analytical Rigour (10 points)

Criteria	Excellent (9-10 points)	Good (7-8 points)	Satisfactory (5-6 points)	Unsatisfactory (0-4 points)
Analysis & Conclusions	In-depth and precise findings with sound conclusions.	Good analysis but could be refined; logical conclusions.	Adequate analysis with some flawed or vague conclusions.	Poor analysis with unsound or unclear conclusions.

Reflection (10 points)

Criteria	Excellent (9-10 points)	Good (7-8 points)	Satisfactory (5-6 points)	Unsatisfactory (0-4 points)
Academic Depth	Exceptionally insightful reflection, backed by academic references.	Good reflection with some academic depth.	Basic reflection with minimal academic underpinning.	Little to no reflection or academic grounding.

Scientific Method (5 points)

Criteria	Excellent (5 points)	Good (4 points)	Satisfactory (3 points)	Unsatisfactory (0-2 points)
Methodology	Rigorous and thorough application of the scientific method.	Good application with minor lapses.	Some application but not consistently applied.	Poor or no application of the scientific method.

Working Files (5 points)

Criteria	Excellent (5 points)	Good (4 points)	Satisfactory (3 points)	Unsatisfactory (0-2 points)
File Submission	All working files are properly submitted and are well-organised.	Most files are submitted and are generally organised.	Some files are missing, or there is a lack of organisation.	Many missing files or complete lack of organisation.
